Escape speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain escape in an ideal gravitational-energy model.

SUCCESS CRITERIA - I can...

1. State the zero-at-infinity energy convention.
2. Distinguish escape speed from circular-orbit speed.
3. Name neglected effects in the ideal calculation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “escape speed” and “orbit” to explain your model or result. Sentence starters are optional.

escape speed: minimum ideal speed to leave without further thrust

orbit: a path around another body under gravity

radius: distance from the center to the edge of a circle or sphere

01 Warm-up: bring back an earlier idea

A 3 kg cart at 10 m/s hits an identical cart at rest. Final velocities are 4 and 6 m/s.
Check momentum and energy; classify the collision.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

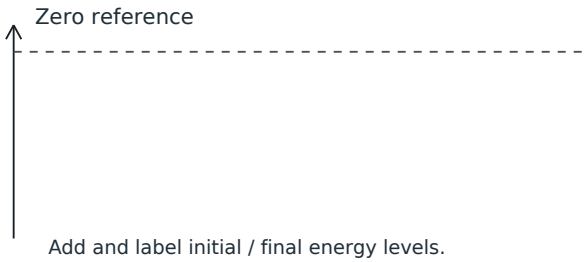
02 Worked example: explain the model

Reference: $v_{escape} = \sqrt{(2\ GM/r)}$; $v_{circular} = \sqrt{(GM/r)}$.
Ideal spherical body; ignore atmosphere/rotation. Minimum escape reaches infinity at zero speed with no further thrust.

Worked reasoning: For a simulated planet with $GM=50\ m^3/s^2$ and radius 4 m, escape speed is $\sqrt{(2\ GM/R)}=5$ m/s. At that limiting speed the ideal final speed at infinity is zero.

Your turn: Draw initial kinetic and gravitational-potential-energy bars whose total is zero.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Add and label initial / final energy levels.

Label K/m, U/m and total E/m in J/kg.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Does escaping mean that gravity turns off at the launch point?
Explain what the speed allows in an energy model.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

04 Apply the physics

For GM=400 m^3/s^2 and r=8 m, calculate circular speed and escape speed.
Compare their ratio.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

Why is an ideal escape-speed calculation not a complete rocket-launch plan?

Word help: ideal = using stated simplifying conditions

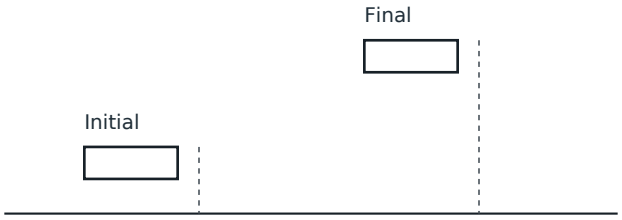
Sentence starter: My design shows ... . It meets the requirement by ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 6 kg object descends 8 m near Earth.
Use g=10.
Find ΔU and the work done by gravity.
Label the distances or energy levels used in your reasoning.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Choose and label a zero-height level.

Marzano check - circle: 1 2 3

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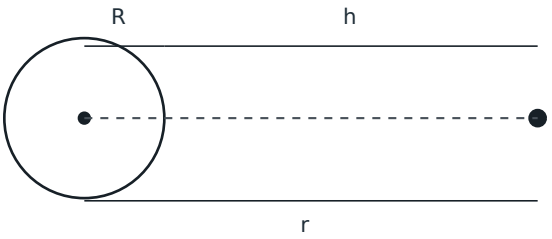
Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

At fixed GM, launch distance increases by factor 4.
Compare escape speeds.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

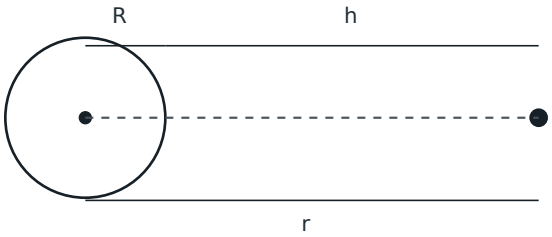


Marzano check - circle: 1 2 3

08 STRETCH: test the model

A planet's mass doubles while radius stays fixed.
Compare escape speed. Then double both mass and radius and compare again.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3